Algorithm

1.Start

2.Input the order n of the square matrices.

3.Input elements of Matrix A and Matrix B.

4.Create a result matrix R of size n × n and initialize all elements to 0.

5.Initialize loop variable i from 0 to n−1.

6.Inside that, initialize loop variable j from 0 to n−1.

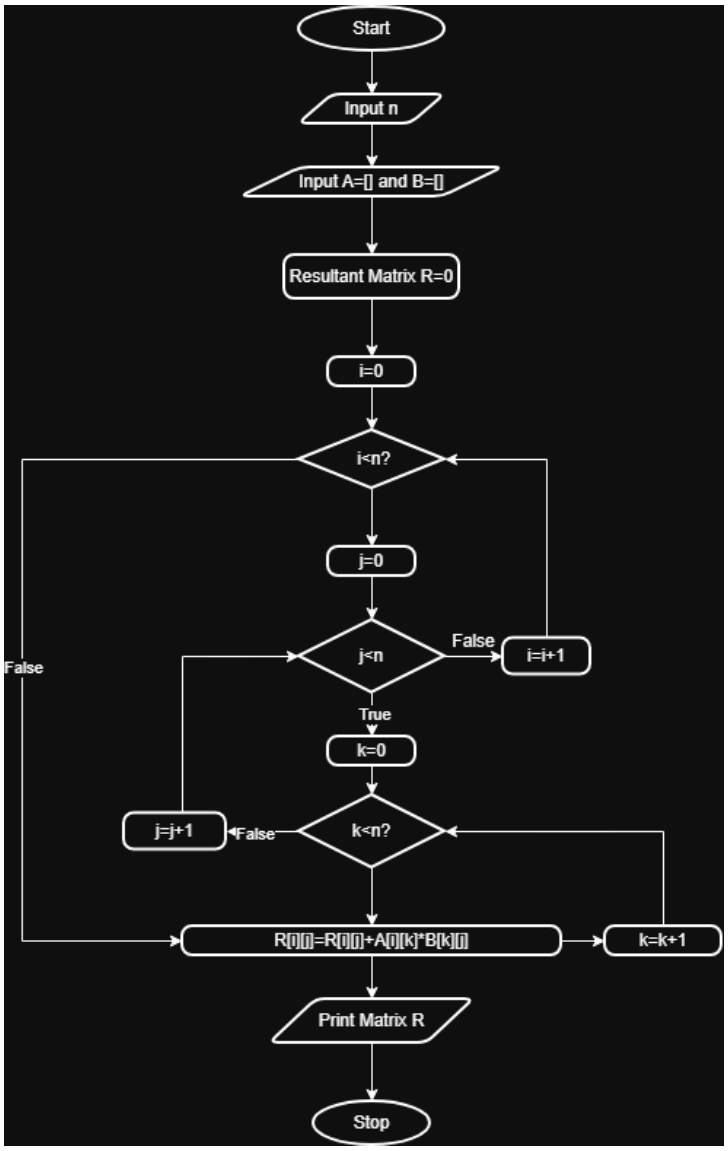
7.Inside that, initialize loop variable k from 0 to n−1.

8.Perform multiplication and addition: If inside loops → R[i][j] = R[i][j] + A[i][k] × B[k][j]

9.Repeat until all loops finish.

10.Print the result matrix R.

11.Stop.



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8.1.1. Multiplication of Two Square Matrices

00:20

Write a Python program to perform the multiplication of two square matrices. The program should take two matrices as input and output their product.

Input Format:

- The first line prints the "dimension: " and reads an integer n, representing the dimensions of the square matrices ($n \times n$).
- The second line prints: "first matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the first matrix.
- Next line prints: "second matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the second matrix.

Output Format:

- Print "Resultant Matrix:" followed by n lines, each containing n integers separated by spaces, representing the product of the two matrices.

Sample Test Cases +

Explorer

Multiplica...

Submit

```
n=int(input("dimension: "))
print("first matrix:")
A=[]
for i in range(n):
    row=list(map(int,input().split()))
    A.append(row)
print("second matrix:")
B=[]
for i in range(n):
```

Average time 0.015 s 14.50 ms

Maximum time 0.017 s 17.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 12 ms Debug

Expected output	Actual output
dimension: 2	dimension: 2
first matrix:	first matrix:
1 2	1 2
3 4	3 4
second matrix:	second matrix:
5 6	5 6

Terminal Test cases

Debugger

Prev Reset Submit Next